



OptiZone Startup Project: Workload Data Processing and Visualization Pipeline for ACL Injury Prevention in Women's Football

EPFL Semester Project | MSc. Robotics Engineering

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Semester: Fall 2025

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January 8, 2026

Abstract & Acknowledgements

This project was conducted in collaboration with the OptiZone startup and aims to contribute to the prevention of anterior cruciate ligament (ACL) injuries in women's football through workload monitoring and data visualization. Using Global Positioning System (GPS) and Heart Rate (HR) data recorded during a football match in Geneva, a complete data processing and visualization pipeline was developed. The system cleans and processes raw wearable data, computes relevant workload and intensity metrics, and produces modular, coach-oriented visualizations. Particular attention was given to code reusability, configurability, and deployability, enabling future integration into server-based platforms. This report documents the scientific background, technical design, and practical applications of the developed solution, highlighting its relevance for performance optimization and injury prevention.

I would like to sincerely thank the OptiZone startup team for their trust, guidance, and continuous feedback throughout this project. Their domain expertise in sports performance and injury prevention strongly shaped the direction and applicability of this work. I also express my gratitude to my EPFL supervising professor for academic guidance, constructive feedback, and support during the semester.

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Chapter 1

Introduction

Anterior cruciate ligament (ACL) injuries represent one of the most severe and frequent injuries in women’s football, often leading to long recovery periods and long-term performance limitations [1]. Recent advances in wearable technologies, such as GPS trackers and heart rate monitors, have enabled detailed monitoring of athletes’ physical load during training and competition. Leveraging such data offers promising opportunities to better understand fatigue, workload accumulation, and injury risk factors.

OptiZone is a sports technology startup whose mission is to predict and prevent ACL injuries in female football players by monitoring fatigue and workload through data-driven approaches. In this context, the objective of this semester project was to design and implement a modular, reusable software pipeline capable of transforming raw GPS and heart rate data into actionable insights for coaches and practitioners.

The project focuses on a real match dataset recorded in Geneva and aims to:

- Clean and preprocess raw GPS and HR data.
- Compute relevant workload and intensity metrics.
- Visualize metrics in a coach-friendly and interpretable manner.
- Provide a configurable and reusable codebase suitable for deployment.

This report presents the scientific background, technical design choices, data processing methodology, computed workload metrics, and visualization strategies, before discussing their implications for performance optimization and injury prevention.

Chapter 2

Background & Sports Science Research

2.1 ACL Injuries in Women's Football

Women athletes exhibit a significantly higher risk of ACL injuries compared to their male counterparts, due to a combination of biomechanical, hormonal, and neuromuscular factors [2]. Match congestion, accumulated fatigue, and sudden spikes in workload are recognized contributors to injury risk.

2.2 External and Internal Load Monitoring

Workload monitoring is commonly divided into two different categories:

- **External load:** mechanical and locomotor demands measured using GPS data, such as distance covered, speed, and accelerations.
- **Internal load:** physiological responses to external demands, often measured using heart rate data [3] or physiological tests.

Combining both perspectives allows practitioners to better interpret athlete responses to training and competition demands.

2.3 Workload Metrics in Practice

Common workload indicators include total distance, time spent in speed zones, number of high-intensity accelerations, and heart rate intensity distributions [4]. These metrics are widely used due to their interpretability and relevance for training load management. In order to find the optimal workload management for an athlete, the idea could be to follow a method close to those shown in the figures below:

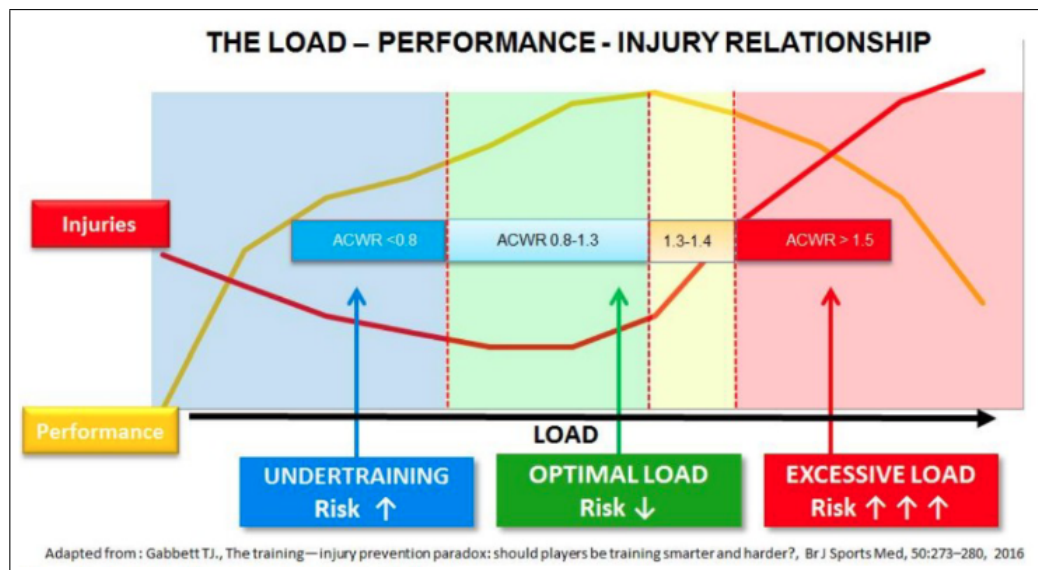


Figure 2.1: Conceptual illustration of the relationship between load - Performance - Injury

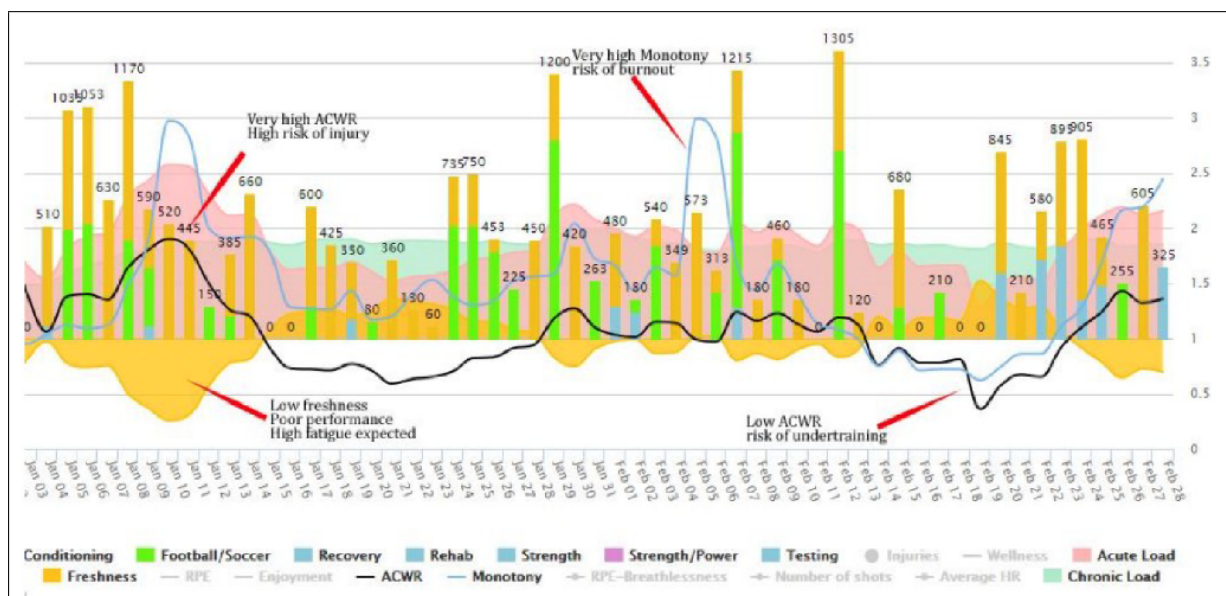


Figure 2.2: Conceptual illustration of the workload management dynamics

Chapter 3

System Overview & Design Goals

The developed system follows a modular pipeline architecture designed to transform raw wearable data into processed workload metrics and visual outputs. The pipeline starts from a CSV file containing GPS and Heart Rate measurements and ends with a set of plots and summary metrics intended for coaches and performance staff.

Several key design goals guided the development of the system:

- **Modularity:** Each processing step is encapsulated in an independent module, allowing easy modification or extension.
- **Reusability:** The codebase is designed to be reused across athletes, sessions, and future projects.
- **Configurability:** User preferences and thresholds are controlled via an easy-to-understand configuration file "config_params.ini", without requiring code modification.
- **Deployability:** The architecture is compatible with server-side execution for integration into larger platforms.

The overall architecture consists of data ingestion and cleaning, metric computation, and visualization modules, orchestrated by a main execution script "optizone_report.py".

Chapter 4

Repository Structure & Code Usage

4.1 Repository Layout

The repository is structured as follows:

- `src/`: main source code.
- `csv_handler`: data loading and cleaning utilities.
- `metrics_handler`: workload metric computation module.
- `plot_handler`: visualization report generators.
- `config_handler`: configuration parsing and loading into main code.
- `config_params.ini`: user-defined parameters.
- `data_samples/`: example datasets of athlete's recorded data.

4.2 Installation and Execution

The project requires Python 3 and standard scientific libraries. The GitHub code repository can be found at: <https://github.com/Mix-ABELA/OptiZone-Data-Visualisation.git>. Dependencies can be installed using:

```
pip install -r requirements.txt
```

The main pipeline is executed via the `optizone_report.py` script.

4.3 Data Requirements

The input dataset must contain timestamped GPS coordinates, speed-related fields, heart rate data, and signal quality indicators. The example CSV file in the `dataset_samples` repository shows the required input data for the visualization report to work

Chapter 5

Configuration System

To ensure flexibility and ease of reuse, the pipeline is entirely driven by a user-editable configuration file named `config_params.ini`. This design choice is particularly relevant for OptiZone’s long-term objective of deploying the code on a server and allowing non-technical users (e.g., coaches, practitioners) to adapt the analysis without modifying Python source code.

The configuration file is parsed at runtime using Python’s `configparser` module, and its values are propagated through the pipeline to control:

- which dataset is processed and which results are displayed,
- which GPS samples are considered valid (data quality filtering),
- athlete-specific physiological constants used for HR/fatigue-related computations,
- the mapping of GPS coordinates to the pitch model,
- the speed thresholds used to classify movement intensities and compute workload distributions.

Unless otherwise specified:

- all YES/NO toggles must be written exactly as YES or NO (case-sensitive),
- all thresholds are floating-point values,
- speed thresholds are expressed in **m/s**,
- GPS coordinates are expressed in **latitude/longitude (WGS84)**.

5.1 Section: [GENERAL REPORT SETTINGS]

This section controls the high-level behavior of the report/pipeline: which dataset is loaded and which families of outputs are computed/displayed.

Table 5.1: Parameters in [GENERAL REPORT SETTINGS].

Parameter	Type / Values	Meaning and effect
Dataset_File_Name	string (filename)	Name of the CSV file to load from the /data_samples directory (or the dataset directory configured in the main script). Example: GPS_Sample.csv.
show_GPS_metrics	YES/NO	If YES, GPS-derived workload metrics are computed and printed (e.g., distance, speed zones, accelerations) and related plots are generated. If NO, GPS outputs are skipped.
show_HR_metrics	YES/NO	If YES, heart-rate-based metrics are computed and printed (e.g., average HR, HR zones, TRIMP) and related plots are generated. If NO, HR outputs are skipped.
show_Fitness_metrics	YES/NO	If YES, fitness/efficiency indicators are computed (e.g., aerobic efficiency, cardiac cost as implemented in the metrics module). If NO, those outputs are skipped.
show_Pitch_Positions	YES/NO	If YES, the athlete's position trace on the pitch is displayed (trajectory plot). This requires valid pitch coordinates in [PITCH GPS COORDINATES].
show_Pitch_Heatmap	YES/NO	If YES, a pitch heatmap is generated to visualize spatial density of movement. Also depends on correct pitch corner coordinates.

Practical note: In a deployment context, these toggles allow generating a lightweight output (e.g., HR-only) when GPS is missing, or a full report when both data streams are available.

5.2 Section: [CSV DATA PROCESSING]

Wearable GPS data can contain unreliable samples due to poor satellite reception, multipath effects, or temporary signal losses. This section defines **data quality filters** applied before computing any metrics, preventing artificial inflation of workload indicators (notably acceleration-based ones).

Table 5.2: Parameters in [CSV DATA PROCESSING].

Parameter	Type Units	/	Meaning and effect
<code>min_Number_Satellites</code>	integer		Minimum number of satellites required for a GPS sample to be considered reliable. Samples recorded with fewer satellites are discarded during preprocessing. Higher values generally yield cleaner trajectories but may remove more data.
<code>min_Quality_of_Signal</code>	numeric (device-specific scale)		Minimum signal quality threshold. Samples below this value are removed. This parameter is device-dependent: its interpretation (e.g., HDOP-like scale, quality score) depends on the GPS tracker manufacturer/export.

Why it matters: Even a few erroneous points can create unrealistic jumps in position, which then produce extremely high speed/acceleration estimates. By filtering low-quality samples first, the pipeline ensures that computed workload metrics remain physiologically meaningful.

5.3 Section: [ATHLETE PERSONAL INFORMATION]

Internal load metrics and intensity interpretations often require athlete-specific baseline information. This section defines physiological constants used by the HR / fitness computation logic.

Table 5.3: Parameters in [ATHLETE PERSONAL INFORMATION].

Parameter	Type Units	/	Meaning and effect
<code>Athlete_Age</code>	integer (years)		Athlete age in years. Used to compute reference values (e.g., estimated maximal HR when such modeling is used by the HR metrics module).
<code>Resting_HR</code>	integer (BPM)		Resting heart rate in beats per minute. Used to compute reserve heart rate and relative intensity proxies. This improves inter-athlete comparability when interpreting HR stress.

Practical note: These values are ideally updated per athlete profile in a deployed OptiZone

system. In a multi-athlete workflow, the configuration could be generated automatically from a roster database.

5.4 Section: [PITCH GPS COORDINATES]

To produce pitch visualizations (trajectory map and heatmap), raw GPS latitude/longitude coordinates must be mapped into a *pitch reference frame*. This section defines the corner coordinates of the pitch used to build that mapping.

Table 5.4: Parameters in [PITCH GPS COORDINATES].

Parameter	Type Units	/ Meaning and effect
Lat_Bottom_Left	float (deg)	Latitude of the bottom-left corner of the pitch.
Lon_Bottom_Left	float (deg)	Longitude of the bottom-left corner of the pitch.
Lat_Bottom_Right	float (deg)	Latitude of the bottom-right corner of the pitch.
Lon_Bottom_Right	float (deg)	Longitude of the bottom-right corner of the pitch.
Lat_Top_Right	float (deg)	Latitude of the top-right corner of the pitch.
Lon_Top_Right	float (deg)	Longitude of the top-right corner of the pitch.
Lat_Top_Left	float (deg)	Latitude of the top-left corner of the pitch.
Lon_Top_Left	float (deg)	Longitude of the top-left corner of the pitch.

How to set these values:

- The recommended approach is to obtain corner coordinates directly from the GPS dataset if the device logs reference markers, or by measuring/estimating pitch corners (e.g., using a GPS-enabled map tool).
- Incorrect or inconsistent corner coordinates will lead to distorted pitch plots (stretched/rotated heatmaps or trajectories).

Why this section is separated: In practice, pitch coordinates vary between stadiums. Keeping them in the configuration file enables fast adaptation of the visualization outputs to new match locations.

5.5 Section: [MOVING INTENSITY THRESHOLDS]

Workload metrics are often reported in terms of movement intensity zones (e.g., walking, jogging, sprinting). This section defines the speed thresholds (in **m/s**) used to classify each time sample

into an intensity zone. These thresholds directly affect distributions such as time spent in zones and can influence training recommendations.

Table 5.5: Parameters in [MOVING INTENSITY THRESHOLDS].

Parameter	Type Units	/ Meaning and effect
Stopped_speed_threshold	float (m/s)	Maximum speed still considered as “stopped” or negligible movement. Samples below this are classified as stopped.
Walking_speed_threshold	float (m/s)	Upper threshold of walking. Speeds between stopped and walking threshold are classified as walking (depending on implementation order).
Jogging_speed_threshold	float (m/s)	Upper threshold of jogging. Determines the range classified as jogging.
Running_speed_threshold	float (m/s)	Upper threshold of running. Speeds between jogging and this threshold are classified as running.
Sprinting_speed_threshold	float (m/s)	Threshold for sprinting classification (typically the highest zone). Speeds above this threshold are counted as sprinting.

Interpretation and tuning:

- These values should be adapted to the population (youth vs elite) and potentially individualized.
- Changing thresholds alters the resulting speed zone percentages, distance in zones, and any “high-intensity” summaries derived from those zones.

Chapter 6

Data Cleaning & Processing Methodology

Wearable GPS and heart rate data recorded during football matches are inherently affected by noise, missing samples, and sensor-related artifacts. If left untreated, these imperfections can significantly bias derived kinematic variables such as speed and acceleration, which in turn directly affect workload metrics. Consequently, a robust and transparent data cleaning pipeline is a critical prerequisite for any reliable workload analysis.

The data processing methodology implemented in this project follows a sequential approach: raw data validation, quality-based filtering, temporal consistency checks, and derivation of kinematic variables. Each step is designed to minimize the influence of measurement noise while preserving physiologically meaningful movement patterns.

6.1 Raw Data Validation & Parsing

The input dataset is provided as a CSV file containing timestamped GPS and heart rate measurements. Upon loading the dataset, an initial validation step ensures that all required columns are present and correctly formatted. These typically include:

- timestamp or elapsed time,
- latitude and longitude coordinates,
- instantaneous or device-reported speed (if available),
- heart rate values,
- GPS quality indicators (e.g., number of connected satellites, signal quality score).

Invalid rows containing missing timestamps, non-numeric values in key fields, or malformed entries are discarded at this stage. This early validation prevents downstream computation errors and ensures temporal coherence of the dataset.

6.2 Quality-Based GPS Filtering

GPS accuracy is highly dependent on satellite reception conditions. Poor signal quality can result in large position errors, which may appear as sudden jumps in the athlete’s trajectory. To mitigate this issue, a quality-based filtering strategy is applied before any kinematic quantities are computed.

Two complementary quality criteria are used:

- **Minimum number of satellites:** Samples recorded with fewer satellites than the threshold defined in the configuration file are discarded.
- **Minimum signal quality score:** Samples whose signal quality indicator falls below the configured threshold are removed.

These thresholds are fully configurable via the `config_params.ini` file, allowing adaptation to different GPS devices and recording conditions. By removing low-quality samples early in the pipeline, the risk of generating unrealistically high speed or acceleration values is significantly reduced.

6.3 Temporal Consistency and Sampling Assumptions

After quality filtering, the remaining samples are sorted chronologically to ensure strict temporal ordering. The pipeline assumes quasi-uniform sampling intervals, as is typical for wearable GPS devices operating at fixed frequencies (e.g., 5–10 Hz). While minor variations in sampling intervals may occur, large temporal gaps are implicitly handled by excluding samples that cannot support reliable derivative computation.

This assumption allows the use of finite difference methods to estimate kinematic variables while maintaining numerical stability.

6.4 Derivation of Kinematic Variables

Once the cleaned position data are available, kinematic variables are computed. Instantaneous speed is estimated from successive position samples using temporal differentiation of displacement. Acceleration is then obtained as the time derivative of speed.

Mathematically, speed and acceleration are approximated as:

$$v(t_i) = \frac{d(t_i, t_{i-1})}{\Delta t}, \quad a(t_i) = \frac{v(t_i) - v(t_{i-1})}{\Delta t}$$

where $d(t_i, t_{i-1})$ denotes the geodesic distance between consecutive GPS positions and Δt is the elapsed time between samples.

By computing these quantities only after rigorous data cleaning, the resulting kinematic signals reflect realistic movement patterns and form a reliable basis for subsequent workload analysis.

6.5 Pitch Mapping and Coordinate Normalization

For visualization purposes, geographic coordinates (latitude and longitude) are mapped onto a two-dimensional pitch reference frame. This transformation relies on the pitch corner coordinates specified in the configuration file.

The mapping process converts global GPS coordinates into normalized pitch coordinates, enabling the generation of trajectory plots and heatmaps. This step is applied only to cleaned data to ensure that spatial visualizations accurately represent on-field movement patterns.

6.6 Impact on Workload Metrics

The described cleaning and processing steps directly influence the quality of computed workload metrics. In particular:

- Distance and speed zone distributions become robust to GPS jitter.
- Acceleration counts are protected from artificial inflation caused by signal noise.
- Spatial visualizations reflect true positional tendencies rather than measurement artifacts.

By enforcing data quality and physiological plausibility constraints prior to metric computation, the pipeline ensures that the resulting workload indicators can be interpreted with confidence by coaches and performance staff.

Chapter 7

Workload Metrics Computation

Workload metrics aim to quantify the physical and physiological demands imposed on an athlete during a football match. In this project, workload metrics are computed using cleaned GPS and heart rate data and are divided into two main categories: external load metrics and internal load metrics. The selected metrics emphasize interpretability and practical relevance for coaches and practitioners.

All metrics are computed on a per-session basis and can be adapted through configuration parameters, enabling reuse across different athletes, competition levels, and contexts.

7.1 External Load Metrics (GPS-Based)

External load metrics describe the mechanical demands placed on the athlete and are derived from GPS position and speed data after cleaning and filtering.

7.1.1 Total Distance Covered

Total distance covered represents the cumulative distance traveled by the athlete over the entire session. It is computed as the sum of distances between consecutive GPS positions:

$$D_{\text{total}} = \sum_{i=1}^{N-1} d(t_i, t_{i+1})$$

where $d(t_i, t_{i+1})$ denotes the geodesic distance between two consecutive GPS samples.

This metric provides a global indicator of locomotor volume during the match.

7.1.2 Instantaneous Speed

Instantaneous speed is estimated from positional displacement over time:

$$v(t_i) = \frac{d(t_i, t_{i-1})}{\Delta t}$$

Speed is expressed in meters per second and serves as the basis for movement intensity classification and acceleration computation.

7.1.3 Speed Zone Classification

Each speed sample is classified into a movement intensity zone (e.g., stopped, walking, jogging, running, sprinting) based on configurable speed thresholds defined in the configuration file. Time and distance spent in each zone are accumulated over the session.

Speed zone distributions provide coaches with an intuitive breakdown of match intensity and physical demands.

7.1.4 Distance per Speed Zone

For each speed zone, the distance covered while the athlete's speed falls within the corresponding threshold range is computed. This is obtained by summing the distances associated with samples belonging to each zone.

This metric allows differentiation between low-intensity and high-intensity locomotor demands.

7.1.5 Acceleration and Deceleration

Acceleration is computed as the temporal derivative of speed:

$$a(t_i) = \frac{v(t_i) - v(t_{i-1})}{\Delta t}$$

Positive values correspond to accelerations, while negative values correspond to decelerations. Only acceleration values exceeding predefined thresholds are considered to avoid counting noise-induced fluctuations.

Acceleration and deceleration metrics are particularly relevant as they reflect high mechanical stress actions associated with neuromuscular fatigue and injury risk.

7.1.6 High-Intensity Actions

High-intensity actions are identified as samples where either speed or acceleration exceeds predefined high-intensity thresholds. The number of such events and their temporal distribution across the match are computed.

These metrics highlight demanding phases of play and potential overload periods.

7.2 Internal Load Metrics (Heart Rate-Based)

Internal load metrics reflect the physiological response of the athlete to the external demands imposed during the match. Heart rate data is used as a proxy for cardiovascular stress.

7.2.1 Average & Maximum Heart Rate

The average heart rate is computed as the mean of all valid heart rate samples recorded during the session. The maximum heart rate corresponds to the highest recorded value.

These metrics provide a general overview of cardiovascular load.

7.2.2 Heart Rate Intensity Distribution

Heart rate samples are categorized into intensity zones based on relative thresholds, which may depend on athlete-specific parameters such as resting heart rate and estimated maximal heart rate. The proportion of time spent in each zone is computed.

Heart rate intensity distributions allow assessment of how demanding the session was relative to the athlete's physiological capacity.

7.2.3 Relative Intensity Indicators

Using resting heart rate and age-based estimates, relative intensity indicators are computed to contextualize absolute heart rate values. These indicators improve inter-athlete comparability and provide insight into physiological strain.

7.3 Fitness and Efficiency Indicators

When enabled, the workload metrics module computes simple fitness-related indicators that relate external and internal load. For example, the relationship between distance covered and average heart rate can be used as a proxy for aerobic efficiency.

Such indicators are exploratory and intended to support qualitative interpretation rather than serve as standalone diagnostic tools.

7.4 Temporal Aggregation and Reporting

All workload metrics can be aggregated over the full session or analyzed over specific time windows. This allows identification of intensity variations across match phases, such as early versus late periods.

Computed metrics are passed to the visualization module, where they are presented in a coach-friendly format.

Chapter 8

Visualization Module

The visualization module represents the final stage of the processing pipeline and is responsible for translating computed workload metrics into clear graphical outputs. These visualizations are generated directly from the cleaned dataset and the workload metrics computed in the previous stage. The main objective is to provide coaches and practitioners with an interpretable summary of match demands, while keeping the plots modular, reproducible, and easy to regenerate for any new session or athlete.

All plots are produced using standard Python scientific visualization libraries and are fully driven by configuration parameters (see `config_params.ini`). This allows users to selectively activate GPS metrics plots, heart-rate plots, and pitch-based visualizations depending on their needs and on the availability/quality of the recorded data.

8.1 GPS-Based Visualizations

GPS-based visualizations describe the athlete's **external workload** by representing locomotor intensity, volume, and mechanical demands over time. These plots rely on speed and acceleration values computed from the cleaned GPS signal.

8.1.1 Speed Time-Series Plot

The speed time-series plot is implemented as a continuous line plot representing instantaneous speed as a function of time. Speed values are computed from successive GPS position samples after data cleaning and filtering.

This plot allows practitioners to:

- identify high-intensity phases of play and short bursts of activity,
- observe pacing patterns and intensity variability throughout the match,
- detect potential drops in intensity that may indicate fatigue.

8.1.2 Speed Zone Distribution Pie Chart

Speed zone distributions are visualized using pie charts representing the proportion of time (and/or distance) spent in predefined movement intensity zones. Each GPS sample is classi-

fied into a zone according to configurable thresholds (e.g., stopped, walking, jogging, running, sprinting).

These pie charts provide a compact and coach-friendly summary of match intensity. They allow practitioners to:

- quantify exposure to high-speed running and sprinting,
- compare low-intensity and high-intensity workload contributions,
- link match demands to training prescriptions and recovery planning.

8.2 Heart Rate Visualizations

Heart-rate-based visualizations describe the athlete’s **internal workload**, reflecting the physiological response to the external demands captured by GPS data.

8.2.1 Heart Rate Time-Series Plot

The heart rate time-series plot is generated as a line plot displaying heart rate values as a function of time. This visualization provides a continuous representation of cardiovascular load throughout the match.

It allows practitioners to:

- monitor fluctuations in physiological intensity,
- identify sustained periods of elevated heart rate,
- relate internal stress to locomotor phases observed in the GPS plots.

Line plots are particularly suitable for heart rate data because they emphasize temporal trends and allow qualitative detection of phenomena such as sustained high cardiovascular strain or progressive drift.

8.2.2 Heart Rate Zone Distribution Pie Chart

Heart rate intensity is summarized using pie charts that represent the proportion of time spent in predefined heart rate zones. Zones may be defined relative to athlete-specific reference values such as resting heart rate and estimated maximal heart rate.

These plots complement the GPS speed zone distributions by providing a physiological perspective: two athletes may show similar external loads but very different internal responses, which can indicate differences in fitness, fatigue state, or recovery status.

8.3 Pitch-Based Visualizations

In addition to external and internal workload summaries, the pipeline provides pitch-based visualizations that add **spatial context** to the athlete’s activity. GPS latitude/longitude data

are mapped to a pitch reference frame using the pitch corner coordinates specified in the configuration file. These plots are only generated from cleaned GPS data to ensure reliable spatial representation.

8.3.1 Athlete Position Trajectory on the Pitch

The pitch trajectory visualization displays the athlete's successive positions over the pitch during the match. It is typically represented as a 2D plot where the pitch is drawn as a rectangle and the athlete's movement path is overlaid.

This plot helps coaches and performance staff:

- understand positional tendencies and spatial coverage,
- relate workload metrics to tactical roles (e.g., winger vs central midfielder),
- detect unusual movement patterns (e.g., limited coverage due to fatigue or injury).

8.3.2 Pitch Heatmap

The pitch heatmap represents the spatial density of the athlete's positions during the match. Positions are aggregated over the pitch area (e.g., into a 2D grid), and the resulting occupancy counts are displayed as a heatmap, where higher density regions correspond to areas of frequent presence.

Heatmaps provide an immediate summary of how and where the athlete spent time on the pitch. From a workload interpretation standpoint, this visualization supports:

- identification of the athlete's most occupied zones and role-related demands,
- comparison of spatial workload distribution between halves or match segments,
- contextualization of high workload phases by linking them to field zones.

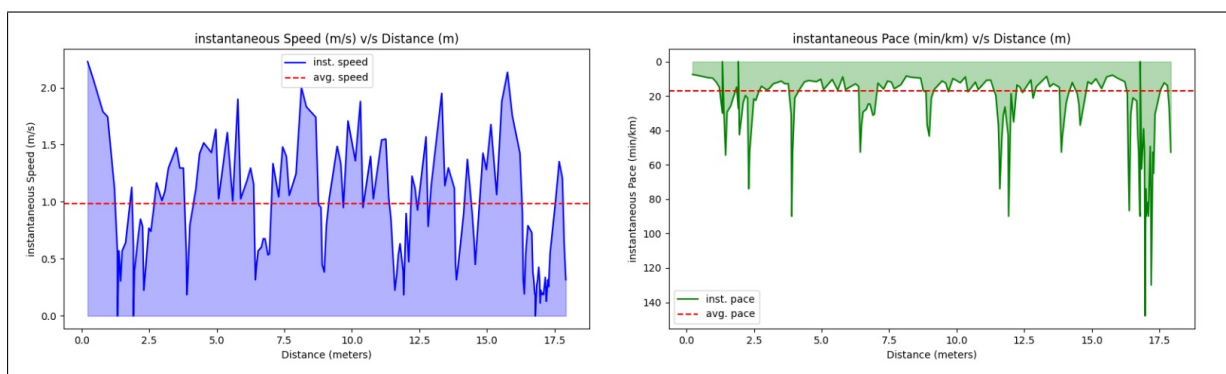


Figure 8.1: Example GPS plots visualization 1

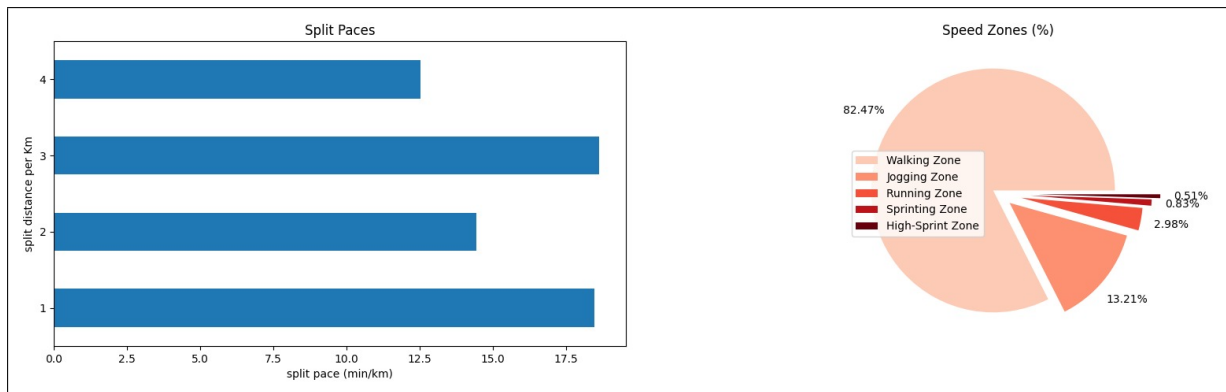


Figure 8.2: Example GPS plots visualization 2

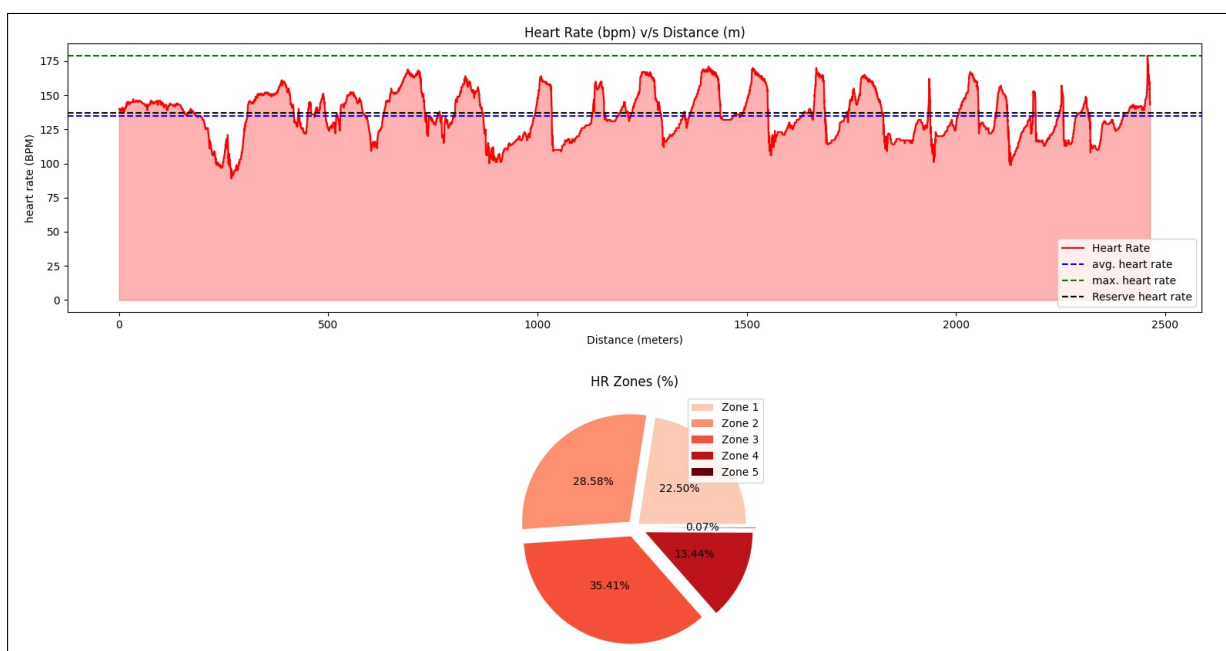


Figure 8.3: Example Heart Rate plots visualization

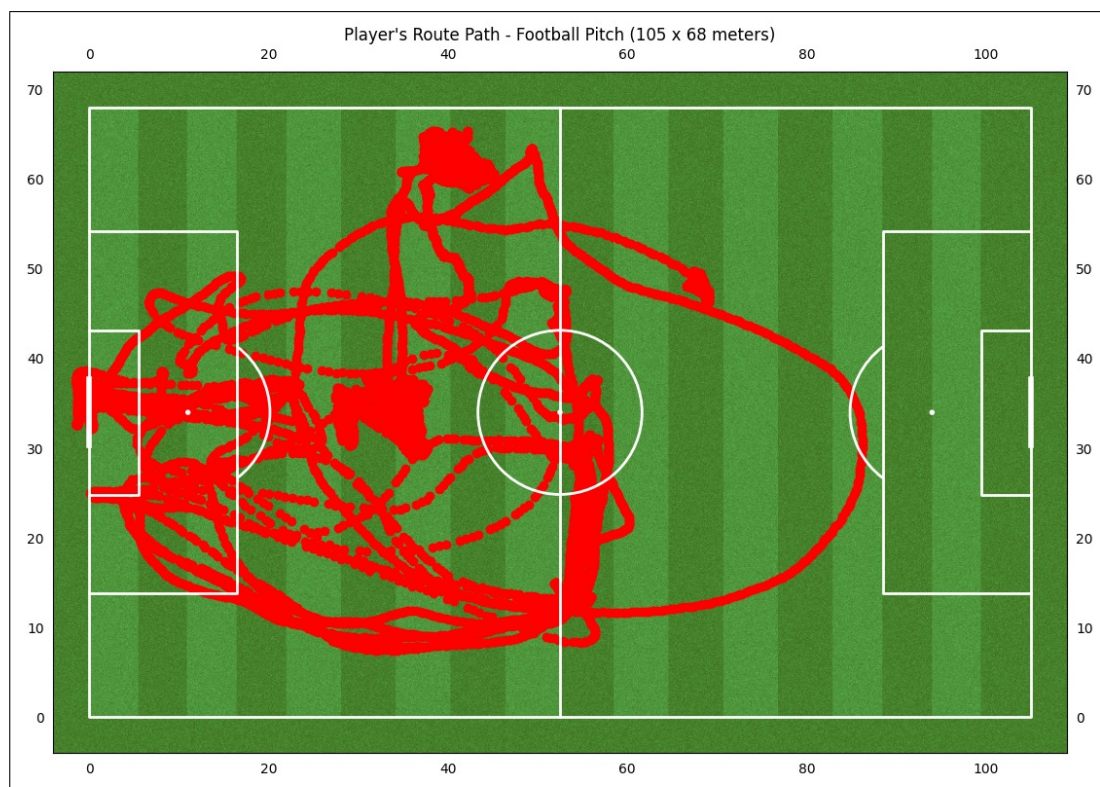


Figure 8.4: Example pitch athlete's positions visualization

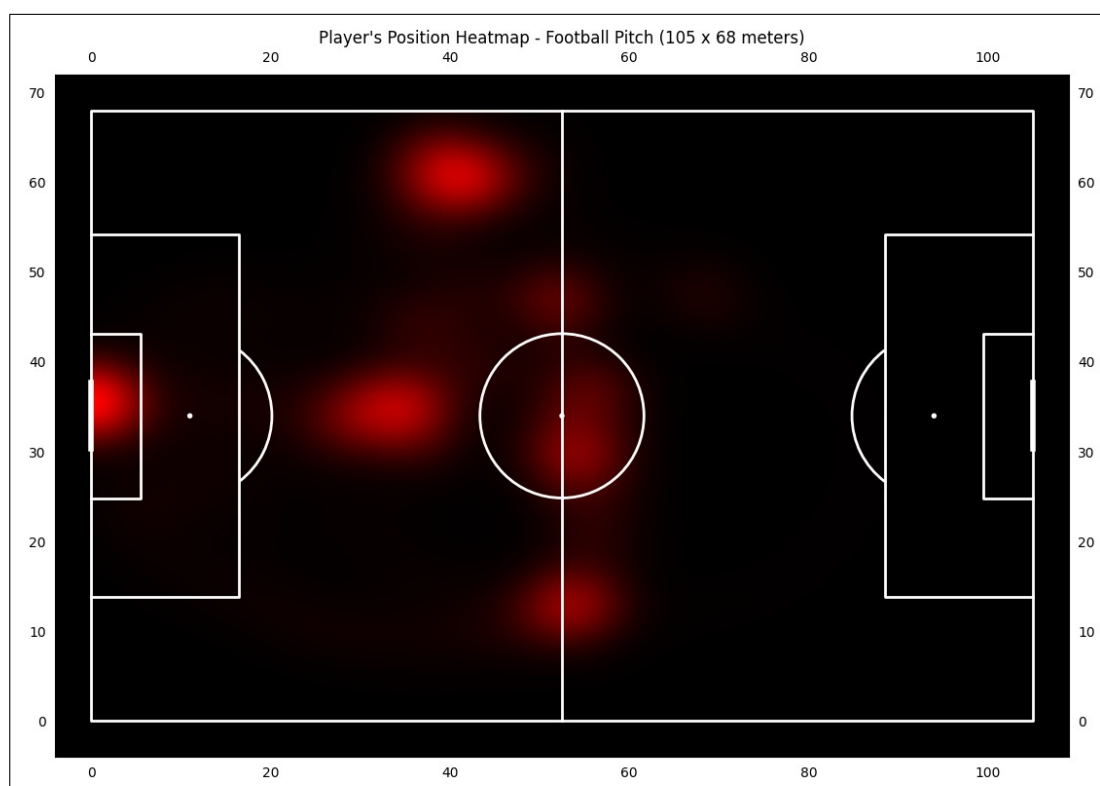


Figure 8.5: Example pitch heatmap positions visualization

Chapter 9

Conclusion

The objective of this semester project was to contribute to the prevention of ACL injuries in women's football by transforming raw wearable data into actionable workload insights. In collaboration with the OptiZone startup, the project addressed a concrete and relevant problem faced by performance and medical staff: how to reliably interpret GPS and heart rate data in order to better understand athlete fatigue, workload exposure, and potential injury risk factors.

Starting from raw GPS and heart rate recordings collected during a competitive football match, the project focused on building a complete and reusable data processing pipeline. A central challenge was the inherent noise and variability of wearable data, which required careful data validation, quality-based filtering, and physiologically informed preprocessing. By implementing a structured cleaning methodology, the project ensured that derived kinematic variables such as speed and acceleration were both reliable and meaningful.

Once the data quality was ensured, the project implemented a set of external and internal workload metrics aligned with applied sports science practices. These metrics were deliberately chosen to balance scientific relevance with practical interpretability, enabling coaches to understand not only the volume of work performed by an athlete, but also its intensity and physiological impact.

A key contribution of this project lies in the design of the visualization module. Rather than focusing solely on numerical outputs, the pipeline emphasizes clear and modular graphical representations of workload metrics. Line plots, pie charts, and distribution plots were selected to reflect how coaches typically interpret performance data, facilitating rapid insight extraction and informed decision-making.

Beyond the analytical components, significant attention was given to software engineering considerations. The modular structure of the codebase, combined with a configuration-driven execution model, ensures that the pipeline can be easily reused, adapted, and deployed in a production environment.

In conclusion, this semester project bridges sports science, data engineering, and applied performance analysis. It demonstrates how a structured and scientifically grounded approach to wearable data processing can support injury prevention strategies while remaining practical and deployable. The developed pipeline provides a solid technical foundation upon which OptiZone can continue to build more advanced analytics, longitudinal monitoring tools, and predictive injury risk models in the future.

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